Recovering leaf attachments from vertex-deleted cards

A path-count certificate for symmetric cores, an infinite family, and an attachment-multiplicity criterion · 2 November 2026

Status: self-contained AI-assisted draft, independent review pending; novelty unassessed. The general Kelly–Ulam reconstruction conjecture remains open. The anchor and endvertex frameworks are established literature, not discoveries claimed here.

1. A precise reconstruction result

Let H be a connected simple graph of order h with minimum degree at least two. Form G by adding two new leaves a,b, adjacent respectively to x,y∈V(H); the two attachment vertices may coincide. Then n=h+2. The deck is the multiset of all n unlabeled one-vertex-deleted induced subgraphs, with repeated cards retained.

Let $p_\ell^H(x,y)$ count vertex-simple x-to-y paths with ℓ edges; $p_0(x,y)=1\{x=y\}$. For x=y and $\ell>0$ the count is zero. The deck determines the attachment signature

$$\Sigma_H(\{x,y\}) = (\text{unordered pair of Aut}(H)\text{ vertex orbits}, p_0^H(x,y),\ldots,p_{h-2}^H(x,y)).$$

Draft certificate theorem. If this signature singles out one unordered-pair orbit under Aut(H), then G is reconstructible among all finite simple graphs with the same full deck. In particular, if the signatures separate every pair orbit of H, all two-leaf extensions of H are reconstructible. The statement does not assume that another reconstruction is already in this class; Section 2 proves that recognition step.

This is stronger than using only attachment distance: different placements can have the same endpoint orbits and distance, but different numbers of simple paths. Section 4 proves the criterion for every clique prism $K_s \square K_2$, s≥3, using only three path counts, despite the core's large automorphism group.

A second criterion in Section 5 handles arbitrary numbers of directly attached leaves by recovering a quadratic moment of their multiplicities. Both are restricted, checkable sufficient conditions. No general reconstruction algorithm with a polynomial-time or uniqueness guarantee is claimed.

2. Recognition from the full deck

More generally, fix H as above and attach $a_v \geq 0$ leaves directly to each core vertex v. Write $r = \Sigma_v a_v \geq 1$ and $G = H(a)$. Every core vertex has degree at least two; exactly the new r vertices have degree one.

Recover which cards delete leaves

For $n \geq 3$, each original edge occurs in $n-2$ cards. Hence the deck recovers the edge count m and the degree of each card's deleted vertex:

$$m = (\Sigma_{C \text{ in deck}} |E(C)|)/(n-2),$$
$$d(C) = m - |E(C)|.$$

The number of cards with $d(C)=1$ is r. These are the leaf-deletion cards. Each such card C consists of H with $r-1$ leaves attached directly to it; deleting all degree-one vertices from C simultaneously exposes H. The remaining leaf counts at the vertices of H form the vector $a - e_v$, considered up to Aut(H). It occurs a_v times in this reduced deck.

Why a deck-mate cannot escape the model

Let J have the same full deck. Its degree multiset is recovered by the same formula, so it also has exactly r leaves. Choose a common card C with $d(C)=1$. Any completion of C to J adds one new vertex of degree one. If that new leaf were attached to one of the $r-1$ existing leaves of C, that old leaf would stop being a leaf, leaving only $r-1$ leaves in J. This contradicts the recovered degree multiset. Thus its neighbor lies in H, and J is also $H(b)$ for some nonnegative vector b with sum r.

This proves recognition against every possible deck-mate. Because core vertices have degree at least two and all added vertices have degree one, an isomorphism $H(a) \cong H(b)$ restricts to an automorphism of H carrying one multiplicity vector to the other. Conversely every such core automorphism extends to an isomorphism of the attached graphs.

$$H(a) \cong H(b) \Leftrightarrow b = \gamma a \text{ for some } \gamma \in \mathrm{Aut}(H).$$

For $r=2$ the two leaf cards reveal the unordered vertex orbits of x,y individually. They do not generally reveal the orbit of the pair. That missing relative placement is exactly the issue addressed next.

This reduction uses an established endvertex/colored-core viewpoint; see Lauri (1993), Section 2. It is restated here with an explicit full-deck recognition argument. The minimum-degree and direct-attachment assumptions are essential to this proof.

3. Recovering the relative placement

Write $c_F(Q)$ for the number of ordinary, not necessarily induced, subgraphs of Q isomorphic to F. In particular P_k denotes a path on k vertices. Extra edges between its vertices are ignored; a triangle contains three copies of P_3.

Kelly counting followed by inclusion–exclusion

Every copy of a k-vertex graph lies in exactly n−k cards. Thus, for k<n,

$$c_F(G) = (1/(n-k))\Sigma_{C \text{ in deck}} c_F(C).$$

For the two-leaf model, let C_a=G−a and C_b=G−b. These two cards, with multiplicity, and the core H are known by Section 2. Inclusion–exclusion gives the number of path copies containing both leaves:

$$\Delta_k = c_{P_k}(G) - c_{P_k}(C_a) - c_{P_k}(C_b) + c_{P_k}(H).$$

Both leaves must be endpoints of any simple path containing them. Removing those endpoints leaves exactly one vertex-simple x-to-y path in H, and the operation is reversible. Therefore, for $0 \le \ell \le h-2$,

$$p_\ell^{H}(x,y) = \Delta_{\ell+3}.$$

The upper limit is exact: $\ell+3 \le h+1 = n-1$. A Hamilton path in H would give a spanning path in G and is not supplied by this counting identity. The zero-length case counts a–x–b when x=y.

Proof of the certificate theorem

Every graph J with the same deck is a two-leaf extension of the same H by Section 2. Its two leaf cards give the same individual vertex orbits, and the displayed identities give the same proper path profile. If that signature has a single pair orbit, the two pairs are carried to each other by a core automorphism. Extending it to the leaves gives J≅G. This proves the theorem.

A finite certificate may list Aut(H), all unordered-pair orbit representatives (including repeated pairs), and their signatures. Full enumeration can be expensive; no polynomial-time bound is asserted. It is not valid to substitute $(A_H^{\ell})_{xy}$: matrix powers count walks that may repeat vertices, whereas this proof uses simple paths.

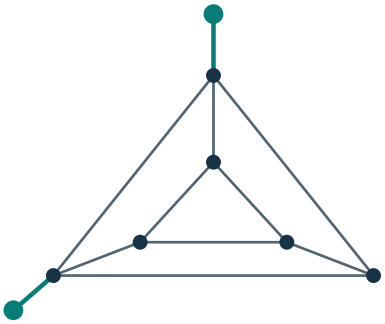
4. An infinite symmetric-core family

Take $H=K_s \square K_2$, $s\geq 3$: two s-cliques joined by a perfect matching. Independent simultaneous permutations of their s positions and exchange of the cliques act transitively on core vertices. They also show that every unordered pair belongs to one of four classes. Their signatures through $\ell=2$ are:

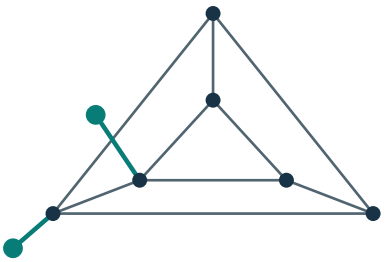
attachment pair	p_0	p_1	p_2
same vertex	1	0	0
distinct, same clique	0	1	$s-2$
matched across cliques	0	1	0
unmatched across cliques	0	0	2

These four signatures are distinct. Within each listed class the stated automorphisms are transitive; between classes the signatures differ, so these are exactly the pair orbits, without needing a separate description of the full automorphism group. Since $h=2s\geq 6$, all required path counts are proper. Section 3 now proves reconstruction of every two-leaf extension of every such core.

For $s=3$, a same-triangle edge and a matching edge are both adjacent pairs in the same vertex orbit. Distance alone cannot distinguish them. Their common-neighbor counts p_2 are one and zero; the reconstructed Δ_5 distinguishes the full graphs. Their two leaf-deleted cards are nevertheless identical as multisets.



same triangle: $p_2=1$



matching edge: $p_2=0$

This is an explicit sufficient-family proof inside a long-studied reconstruction framework. It is not asserted to be a previously unknown reconstructible class, and it is not a proof for arbitrary symmetric cores.

5. Recovering leaf multiplicities

Return to $G=H(a)$ with $r\geq 1$ attached leaves. The leaf cards give the multiset of vectors $b=a-e_v$ up to $\mathrm{Aut}(H)$, each repeated a_v times. The quantity $Q(b)=\Sigma_v b_v^2$ is independent of the chosen core labeling. Summing it over the r cards gives

$$\Sigma_{\text{leaf cards}}\, Q(b) = (r-2)Q(a)+r.$$

Proof. $Q(a-e_v)=Q(a)-2a_v+1$. Multiply by the multiplicity a_v and sum: $rQ(a)-2\Sigma a_v^2+r=(r-2)Q(a)+r$. Thus $Q(a)$ is recovered when $r\neq 2$. In any particular card the removed multiplicity is

$$q = [Q(a)-Q(b)+1]/2.$$

A completion must increment a coordinate whose current value is $q-1$. If this value occurs at a unique core vertex in one leaf card, the vector is restored uniquely, and Section 2 proves full-deck reconstructibility. More generally it is enough that all such vertices form a single orbit under the automorphisms of the occupancy-colored core.

A simple sufficient condition and a uniform bound

Suppose $r\neq 2$ and some positive multiplicity q occurs in a but $q-1$ does not. Delete a leaf from a q-vertex. In the resulting card exactly that vertex has multiplicity $q-1$, so the certificate applies.

Consequently, every $H(a)$ with $r>h(h-1)/2$ is reconstructible by this criterion. To prove this, suppose no positive multiplicity has an absent predecessor and let $M=\max a_v$. Then the values $0,1,\ldots,M$ all occur, so $M\leq h-1$. With h coordinates their largest possible sum is

$$r \leq Mh-M(M+1)/2 \leq h(h-1)/2.$$

The contrapositive gives a gap. Since $h\geq 3$, the displayed strict bound also excludes $r=2$. The vector $(0,1,\ldots,h-1)$ shows that this numeric bound is sharp for forcing this particular gap, not that reconstruction fails at equality.

The quadratic identity is a direct derivation in the established colored-core model. General endvertex reconstruction has stronger group-action tools, including Lauri's 1993 criteria for repeated labels. No priority or improvement over that literature is claimed. The value $r=2$ is singular for this identity; path information from the full deck handles that case in Sections 3–4.

6. Exact limitations of the approach

Two leaf cards alone can lose the whole relative position

For a triangular-prism core all vertices lie in one orbit. Every placement of two leaves therefore has the same leaf-deletion deck: two copies of the core with one leaf. There are four pair orbits in Section 4, giving four nonisomorphic full graphs with that same restricted deck. This is a counterexample to reconstruction from those two cards alone, not to reconstruction from the full vertex deck.

The multiplicity identity reflects precisely this issue: when r=2 its right side is the constant two, independent of Q(a). It cannot recover the missing placement information by division.

No fixed path cutoff works on every core

Fix $L \geq 0$ and take $H = C_{2L+5}$. Attach the two leaves at cycle distances L+1 and L+2 in two separate constructions. Each core is vertex-transitive, so both individual attachment-orbit pairs agree. For every $0 \leq \ell \leq L$ the two path profiles are identically zero: no such short path joins either attachment pair. Their two leaf cards agree, yet the resulting graphs are nonisomorphic because their attachment distances differ.

Thus the two leaf cards plus any fixed cutoff of these path statistics do not identify all large cores. The full proper profile used in the theorem distinguishes these cycle examples. This does not rule out a different bounded family of statistics or a different reconstruction method.

Why the remaining conjecture is still hard

An arbitrary graph need not have any leaves or a core that can be recognized this way. Even in the two-leaf model, the signature-separation hypothesis has not been proved for every core. Finding no collision through seven core vertices is not evidence of a universal theorem. Simple path counting and automorphism enumeration may both be expensive. The framework supplies a certificate where its hypotheses hold and explicit failure modes where they do not.

7. Exact finite checks

The verifier generates unlabeled graph representatives by explicitly marking every permutation orbit of every labeled adjacency mask through order seven. It obtains the standard graph counts 4, 11, 34, 156 and 1,044 at orders 3–7, then keeps exactly the connected minimum-degree-two cores. No spectral equality or floating-point graph comparison is used.

h	eligible cores	pair orbits	separated
3	1	2	2
4	3	10	10
5	11	64	64
6	61	617	617
7	507	8825	8825

All 9,518 unordered-pair orbits across these 583 cores pass the full proper-path certificate. Distance alone leaves 104 of these pair orbits in non-singleton signature classes; the path profile separates them. These counts concern the explicit certificate, not an extension of known finite reconstruction verification.

The program also checks 3,637 Kelly/inclusion–exclusion path identities against direct endpoint-path enumeration, and compares the complete extension decks for all 693 pair orbits over cores through order six. Canonical card comparison enumerates all permutations remaining after deterministic degree refinement; refinement is only a pruning step, not the isomorphism criterion.

Additional checks classify every attachment-multiplicity orbit with one through eight leaves on the triangular prism, verify the norm identity and every reported unique-coordinate restoration, exhaustively check the multiplicity-gap threshold for core sizes three through five, and verify the closed-form clique-prism signatures at clique sizes three through nine.

Reproduce with Python 3.10+ and NumPy: python verify-reconstruction.py. The JSON records exact counts, examples and scope. The tests corroborate examples; the general claims rely on the written proofs. McKay’s existing computation already covers all full graphs through 13 vertices, beyond the sizes in these particular finite checks.

8. Sources, status and the next task

Full conjecture. Every finite simple graph on at least three vertices is determined up to isomorphism by its complete multiset of unlabeled vertex-deleted cards. Contemporary 2026 primary papers still treat this as open. The present contribution does not change that status.

Primary sources checked on 2 November 2026

Kelly counting and context: Dufresne, Jeronimo, Kenkel, Lindo and Villamizar, Shuffling the Deck: Invariant Theory and the Graph Reconstruction Conjecture (2026), Sections 1–3. Primary paper.

Anchor framework: Farhadian, Reconstruction of graphs via asymmetry (2016), Section 6, Theorem 4 and Lemma 5. That paper discusses the relative placement of two vertices outside an anchor and distance-based criteria. This report gives an explicit path-count certificate in a pendant-attachment model. Primary preprint.

Endvertex framework: Josef Lauri, Endvertex-deleted Subgraphs, Ars Combinatoria 36 (1993), 171–182, Section 2. The colored-core reduction and strong group-action criteria precede this report. Primary journal PDF.

Finite benchmark: Brendan McKay, Reconstruction of small graphs and digraphs, version 4 (2022), verifies the full and set conjectures through 13 vertices and specified larger classes. That computation was not rerun here. Primary paper.

Recent structural progress: Heinrich, Kiyomi, Otachi and Schweitzer, Interval Graphs are Reconstructible, version 2 (12 May 2026), proves that class reconstructible with a polynomial-time algorithm. It does not settle arbitrary graphs. Primary paper.

A concrete continuation

Independently audit the recognition argument and every step of the path-count certificate. Then search for a connected minimum-degree-two core of order at least eight with two non-equivalent unordered attachment pairs having the same endpoint orbits and all proper simple-path counts. Such an example would limit this certificate without refuting the full conjecture. If proving separation on a class instead, state the extra hypothesis exactly and compare it with prior anchor and endvertex criteria.

Provenance: user selected graph reconstruction from vertex-deleted subgraphs, continuing shared revision 2026-11-02.1. This is a new direction in the notebook; it does not modify or independently validate the earlier IWN draft. Review: needs-independent-review. Novelty: unassessed. Earlier research records are preserved.